Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Semiconductor diode under no bias, forward bias and reverse bias conditions

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Semiconductor diode under no bias, forward bias and reverse bias conditions

1 Semiconductor diode under no bias, forward bias and reverse bias conditions

Study focus: Semiconductor diode under no bias, forward bias and reverse bias conditions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000010 · EV0000011

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we will go with the topic three, Introduction to Semiconductor.

Key details

- In the last class we have looked at the diode and biasing of the diode, where we know that diode has two regions, P and N type, where P side is connected with the biasing.
- P side is connected to the positive of the terminal and N side is connected to negative of the battery terminal.
- P and N region not connected to the voltage supply.
- We call it as a no bias.
- That means to say, with no voltage applied to the diode, we see that the boundary, there is a depletion region with the charges being accumulated.
- This happens because at the instant the two metals are joined, the electrons and the holes in the region of the junction will combine, resulting in lack of free charges in the region near the junction, as shown, which we call it as a depletion region.
- And note in the figure that only particle displayed in this region are positive and negative ion remaining once the free charges have been observed.
- That means to say, because of the concentration gradient, the charge carriers will start move from.
- P side to N side and N side to P side.
- As the moment they start moving, they forms a charges which is been settled at the boundaries of the P and N, where negative charge is settled at the P region boundary and

positive charge is settled at the N region boundary.
- This region of uncovered positive and negative ion is called depletion region.
- This is due to the depletion of free charge carrier in the region.
- So which we call it as a no bias condition.
- As the name indicates, a forward bias or 'ON' condition.
- The current that exists under the forward bias is called the forward bias current and is represented by ID in the diagram.
- As the name indicates, forward bias, we connect P region to positive terminal of the battery and N region to negative terminal of the battery.
- As the moment we connect it, we call it as an 'ON' condition, that is established by applying a positive potential to the P terminal and a negative potential to the N terminal, as shown in the diagram.
- The application of the forward bias potential that is VD we call here, will pressurize or will pressure the electron in the N-type material and hole in the.
- P-type material to recombine with ion near the boundary and reduces the width of the depletion region, as shown here.
- If compared to the previous diagram, we have a very big depletion region which is shown here.
- And because of the application of the voltage, the depletion region is reduced and we see only a few charge carrier.
- This happens due to the application of the voltage on both P and N region, and which results in the ID current which is a sum of the eye majority and the leakage current.
- This can be represented by a V-A characteristic, as shown, the forward bias current is shown here.
- There is a very less resistance is there in the forward bias and we see that there is a current flow with application of the voltage.
- This is about the forward bias.
- Then what is reverse bias?
- The reverse bias condition, the P side is connected to the negative terminal of the battery, N side is connected to the positive terminal of the battery with applied voltage VD.
- It's increasing due to the large number of free electron drawn from the positive terminal of the applied voltage.
- For a similar reason, the number of uncovered ion will increase in the P-type material.
- The net effect therefore is widening of a depletion region, as shown.
- This widening of the depletion region will establish too a great.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 05:00–09:59

a barrier for majority carrier to overcome, effectively reducing the majority carrier flow to zero.

Key details

- So the current that exists under a reverse bias condition is called the reverse saturation current and is represented as IS.
- And this also represented in the V-A characteristic as shown, with the initial applied voltage because of the depletion region will not be able to see the current flow which has been shown here.
- And once after the reverse breakdown region, once after the voltage called VBR, there is going to be a breakdown with the current increase as shown.
- And this could be realized from a reverse bias condition.
- So today's class we have seen with a no bias condition, forward bias condition and a reverse bias condition.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9939; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...
SRC00007	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...
SRC00008	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...

C Revision and retrieval practice